

The Pop Bottle Magnetometer - Example Data

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Three examples of pop bottle magnetometer plots and comparisons to Bureau of Meteorology geomagnetic data are provided below. Additional information and interpretation of the geomagnetic storm conditions during the recording periods are also included.

In all examples, the pop bottle magnetometer was set up and run in the Physics Building at the University of Tasmania, Hobart. Timelapse videos from the first 2 examples can be found here: [The Pop Bottle Magnetometer: Example Timelapses & Data](#)

Note that the room where the pop bottle magnetometer was placed was not perfectly undisturbed and temperature was not controlled. Both factors will have created unwanted noise in the data sets.

Example 1: 13 – 19/05/2026

A **moderate geomagnetic storm** event was recorded during the observation period (13/05/2026 - 19/05/2026).

The pop bottle magnetometer data shows a clear disturbance on 16/05/2026 at approximately 08:30 Hobart time (22:30 UT on 15/05/2026). This is visible as a spike in the reflected laser spot position reaching 19.4 cm, corresponding to an estimated deflection of $\sim 1^\circ$ in the magnetic field (**Figure 1**).

This disturbance is consistent with a geomagnetic storm event, as the [Kp index](#) reached 6.0 at 21:00 UT on 15/05/2026 (Matzka, 2021). A Kp value of 6 corresponds to a moderate (G2) storm on the [NOAA space weather scale](#).

A disturbance in the [Hobart Magnetometer](#) declination data is consistent with the pop bottle data (**Figure 2**). A spike is clearly visible in the data late on 15/05/2026 (UT) which aligns closely in timing with the disturbance in the pop bottle magnetometer data. This agreement strongly suggests that both instruments recorded the same moderate geomagnetic storm event.

Note: the cotton sewing thread suspending the magnet in the pop bottle magnetometer stretched during this recording period. The laser spot movement over the first day is

mostly due to the thread untwisting and settling. Dental floss was used to hang the magnet in the second two examples, leading to less stretch/ drift.

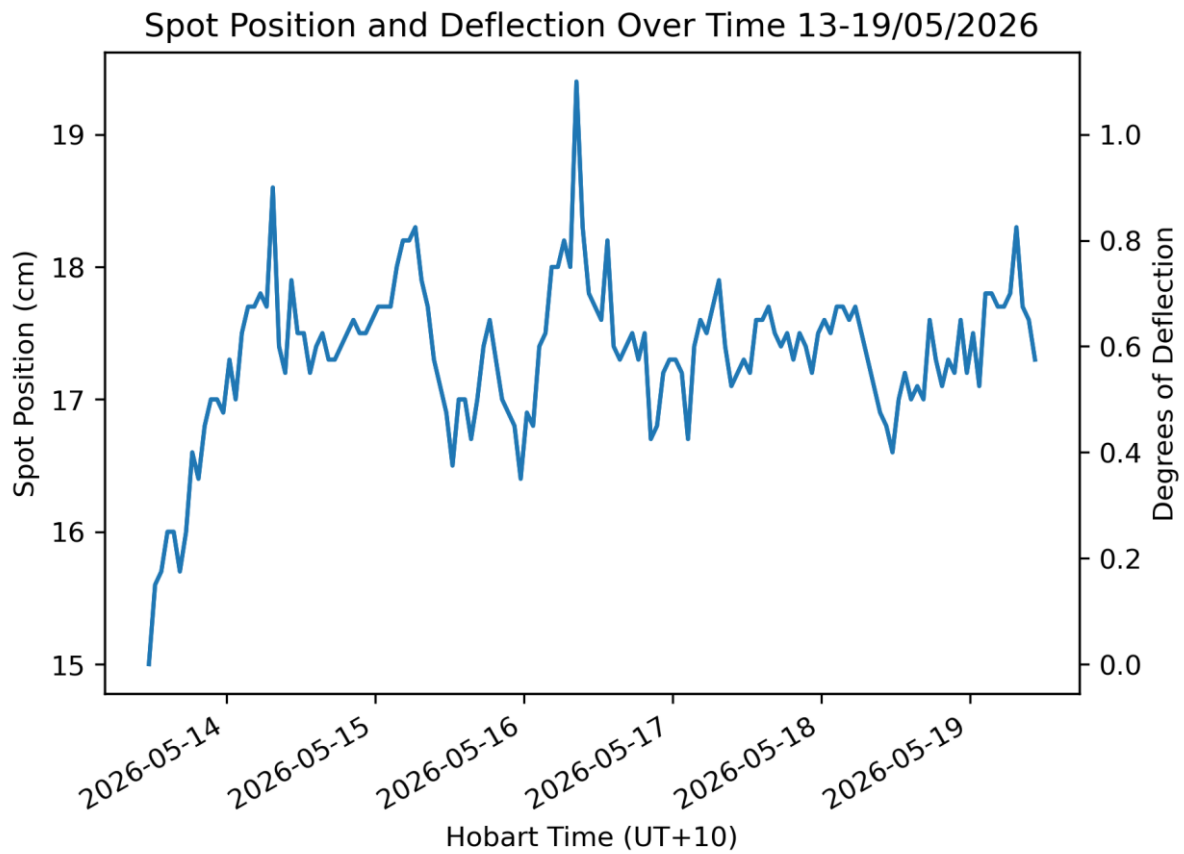


Figure 1: Pop bottle magnetometer data 13-19/05/2026. Hobart time (UT+10). Degrees of geomagnetic deflection was calculated according to NASA (2015). Reference laser spot position was 15 cm.

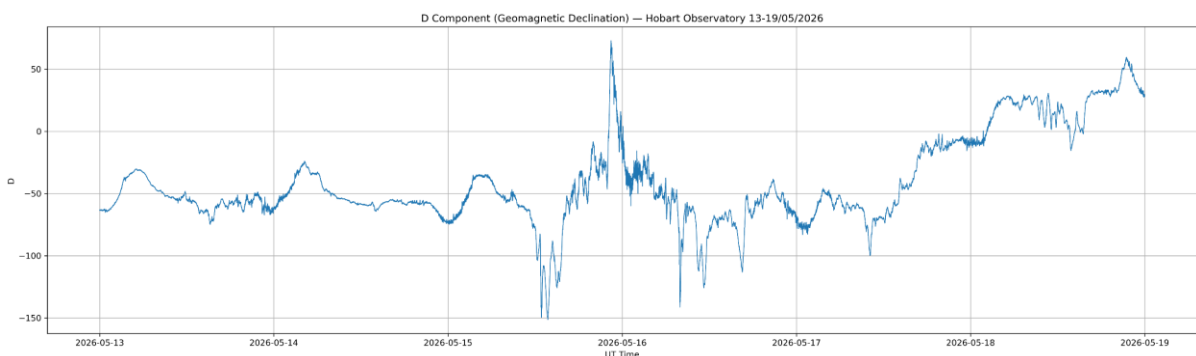


Figure 2: Hobart Magnetometer geomagnetic declination data 13-19/05/2026. UT time. (Space Weather Services (Bureau of Meteorology), World Data Centre for Geomagnetism)

Example 2: 20 – 27/05/2026

Geomagnetic activity levels remained quiet during this observation period (20/05/2026 - 27/05/2026).

No geomagnetic storms occurred during this week as [Kp values](#) remained below 5 throughout (Matzka, 2021). According to [NOAA space weather scales](#), Kp must reach a value of 5 for a minor (G1) storm event to be classified.

This quiet activity is reflected by the small ($< 0.3^\circ$) degrees of deflection and relatively stationary reflected laser spot position (< 1.5 cm displacement) (**Figure 3**). Similarly, the [Hobart Magnetometer](#) declination data (**Figure 4**) lacks any significant disturbances during this period, which is consistent with the pop bottle magnetometer observations.

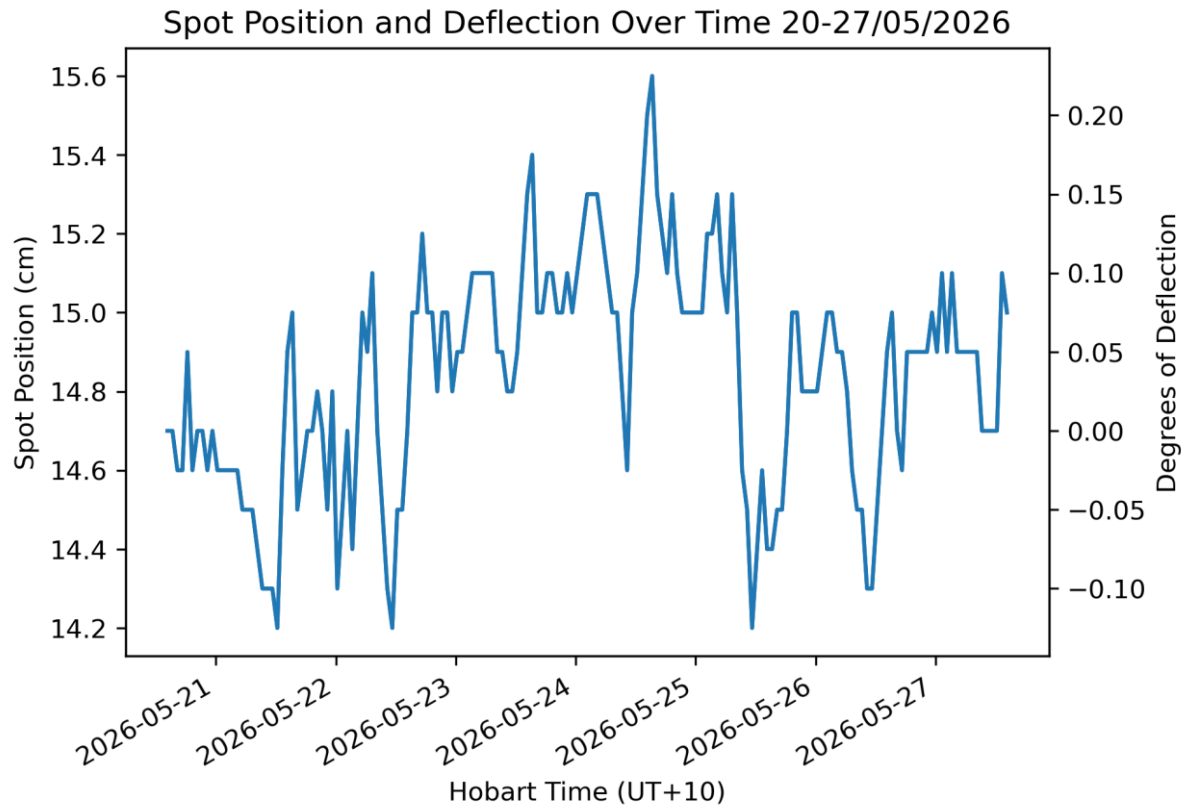


Figure 3: Pop bottle magnetometer data 20-27/05/2026. Hobart time (UT+10). Degrees of geomagnetic deflection was calculated according to NASA (2015). Reference laser spot position was 14.7 cm.

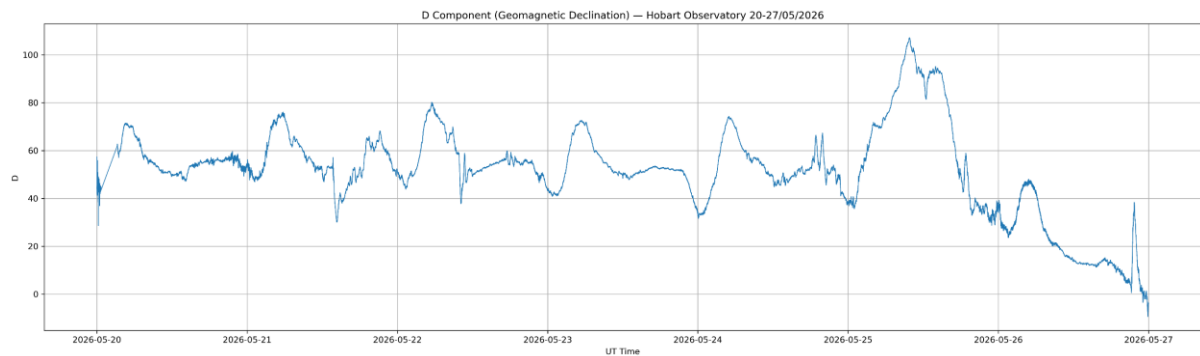


Figure 4: Hobart Magnetometer geomagnetic declination data 20-27/05/2026. UT time. (Space Weather Services (Bureau of Meteorology), World Data Centre for Geomagnetism).

Example 3: 04 – 12/06/2026

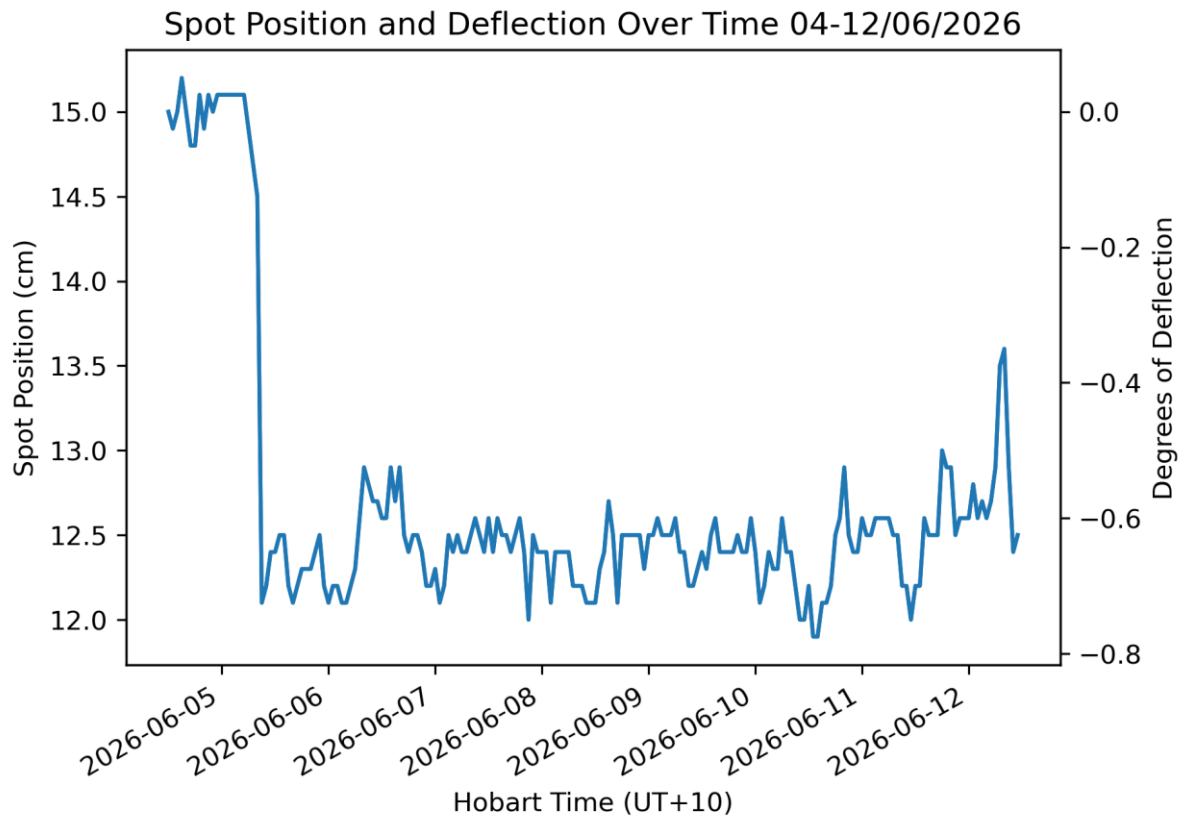


Figure 5: Pop bottle magnetometer data 04-12/06/2026. Hobart time (UT+10). Degrees of geomagnetic deflection was calculated according to NASA (2015). Reference laser spot position was 15 cm.

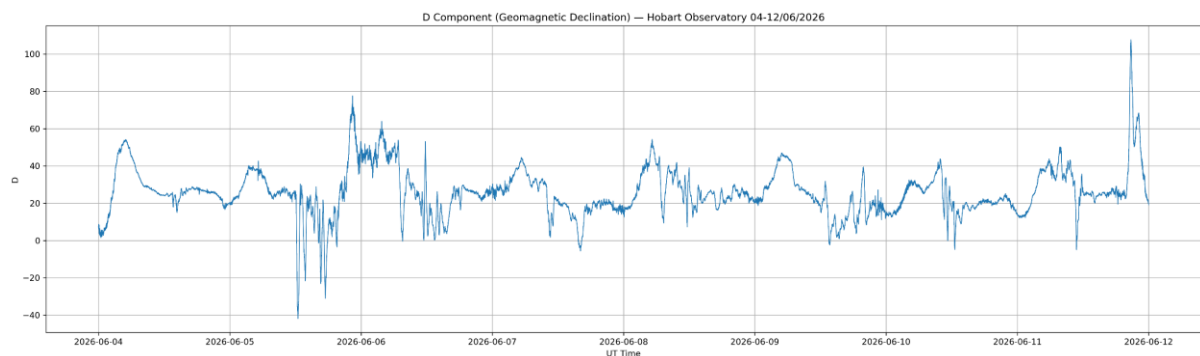


Figure 6: Hobart Magnetometer geomagnetic declination data 04-12/06/2026. UT time. (Space Weather Services (Bureau of Meteorology), World Data Centre for Geomagnetism).

References:

Bureau of Meteorology, Space Weather Services. *Hobart Magnetometer*. Available at: https://sws.bom.gov.au/World_Data_Centre/3/1/35

Bureau of Meteorology, Space Weather Services. *World Data Centre for Geomagnetism: Hobart magnetic observatory data (raw HBT format)*. Available at: https://www.sws.bom.gov.au/World_Data_Centre/1/2

Matzka, J., Stolle, C., Yamazaki, Y., Bronkalla, O. and Morschhauser, A. (2021). The geomagnetic Kp index and derived indices of geomagnetic activity. *Space Weather*. <https://doi.org/10.1029/2020SW002641>

NASA (2015). *Soda Bottle Magnetometers for Space Weather Studies*. Available at: <https://science.nasa.gov/learn/heat/resource/soda-bottle-magnetometers-student-guide/>

NOAA. *NOAA Space Weather Scales*. Available at: <https://www.spaceweather.gov/noaa-scales-explanation>

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